

Laboratory Work 2

Database Normalization and DDL

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Course Databases

1 Normalization

The original Booking_receipt table stores booking, passenger, flight, airline and airport data in one place. This creates repeated data and makes updates harder.

1.1 Normal Form Problems

- 1NF: seat_numbers contains several values in one column. One cell must contain only one value.
- 2NF: after separating the seat numbers, the key is (booking_id, passenger_passport_number). Some columns depend only on booking_id and some depend only on the passport number.
- 3NF: flight details, airline name and airport data depend on other non-key columns, not directly on booking_id.

1.2 Step-by-Step Result

Primary keys are underlined. Foreign keys are marked as FK.

First Normal Form

Table	Attributes
Booking_receipt_1NF	<u>booking_id</u> [PK], <u>passenger_passport_number</u> [PK], seat_number, passenger_first_name, passenger_last_name, flight_id, airline_name, departure_airport_name, departure_city, arrival_airport_name, arrival_city, departure_time, arrival_time, booking_date, amount, payment_method

Second Normal Form

Table	Attributes
Booking_2NF	<u>booking_id</u> [PK], flight_id, airline_name, departure_airport_name, departure_city, arrival_airport_name, arrival_city, departure_time, arrival_time, booking_date, amount, payment_method
Passenger	<u>passport_number</u> [PK], first_name, last_name
Booking_passenger	<u>booking_id</u> [PK, FK], <u>passport_number</u> [PK, FK], seat_number

Third Normal Form

Table	Attributes
Airline	<u>airline_id</u> [PK], airline_name
Airport	<u>airport_id</u> [PK], airport_name, city
Flight	<u>flight_id</u> [PK], airline_id [FK], departing_airport_id [FK], arriving_airport_id [FK], departure_time, arrival_time
Booking	<u>booking_id</u> [PK], flight_id [FK], booking_date, amount, payment_method
Passenger	<u>passport_number</u> [PK], first_name, last_name
Booking_passenger	<u>booking_id</u> [PK, FK], <u>passport_number</u> [PK, FK], seat_number

1.3 Functional Dependencies

- booking_id → flight_id, booking_date, amount, payment_method
- passport_number → first_name, last_name
- (booking_id, passport_number) → seat_number
- flight_id → airline_id, departing_airport_id, arriving_airport_id, departure_time, arrival_time
- airline_id → airline_name
- airport_id → airport_name, city

1.4 Flights Table

The Flights table is in 3NF if flight_id is the primary key. All flight columns depend on flight_id, while airline_id and airport IDs are only foreign keys. Airline and airport descriptions are stored in separate tables.

2 SQL for the 3NF Tables

The schema below separates the data into six related tables.

```
CREATE SCHEMA IF NOT EXISTS lab2_norm;
SET search_path TO lab2_norm;

CREATE TABLE airline (
    airline_id CHAR(2) PRIMARY KEY,
    airline_name VARCHAR(50) NOT NULL
);

CREATE TABLE airport (
    airport_id INT PRIMARY KEY,
    airport_name VARCHAR(80) NOT NULL,
    city VARCHAR(50) NOT NULL
);

CREATE TABLE flight (
    flight_id VARCHAR(10) PRIMARY KEY,
    airline_id CHAR(2) NOT NULL REFERENCES airline(airline_id),
    departing_airport_id INT NOT NULL REFERENCES airport(airport_id),
    arriving_airport_id INT NOT NULL REFERENCES airport(airport_id),
    departure_time TIMESTAMP NOT NULL,
    arrival_time TIMESTAMP NOT NULL
);

CREATE TABLE passenger (
    passport_number VARCHAR(20) PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL
);

CREATE TABLE booking (
    booking_id INT PRIMARY KEY,
    flight_id VARCHAR(10) NOT NULL REFERENCES flight(flight_id),
    booking_date DATE NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_method VARCHAR(30) NOT NULL
);

CREATE TABLE booking_passenger (
    booking_id INT NOT NULL REFERENCES booking(booking_id),
    passport_number VARCHAR(20) NOT NULL REFERENCES passenger(passport_number),
    seat_number VARCHAR(10) NOT NULL,
    PRIMARY KEY (booking_id, passport_number)
);
```

2.1 Sample Data

```
INSERT INTO airline VALUES
('KC', 'Air Astana'), ('LH', 'Lufthansa'), ('TK', 'Turkish Airlines'),
('QR', 'Qatar Airways'), ('EK', 'Emirates');

INSERT INTO airport VALUES
(1, 'Almaty International Airport', 'Almaty'),
(2, 'Astana International Airport', 'Astana'),
(3, 'Frankfurt Airport', 'Frankfurt'),
(4, 'Istanbul Airport', 'Istanbul'),
(5, 'Hamad International Airport', 'Doha');

INSERT INTO flight VALUES
('KC101', 'KC', 1, 2, '2026-10-01 08:00', '2026-10-01 09:40'),
('KC901', 'KC', 1, 3, '2026-10-02 06:30', '2026-10-02 11:10'),
('TK351', 'TK', 2, 4, '2026-10-03 09:00', '2026-10-03 12:30'),
('QR390', 'QR', 1, 5, '2026-10-04 04:30', '2026-10-04 07:20'),
('LH648', 'LH', 3, 1, '2026-10-05 13:00', '2026-10-05 22:30');

INSERT INTO passenger VALUES
('N1234567', 'Aruzhan', 'Sarsenova'),
('N2345678', 'Dias', 'Omarov'),
('N3456789', 'Amina', 'Bekova');
```

```
('N4567890', 'Timur', 'Aliyev'),
('N5678901', 'Dana', 'Kim');

INSERT INTO booking VALUES
(1001, 'KC101', '2026-09-10', 45000.00, 'Card'),
(1002, 'KC901', '2026-09-11', 185000.00, 'Card'),
(1003, 'TK351', '2026-09-12', 140000.00, 'Cash'),
(1004, 'QR390', '2026-09-13', 210000.00, 'Card'),
(1005, 'LH648', '2026-09-14', 225000.00, 'Card');

INSERT INTO booking_passenger VALUES
(1001, 'N1234567', '12A'), (1002, 'N2345678', '18C'),
(1003, 'N3456789', '7B'), (1004, 'N4567890', '22D'),
(1005, 'N5678901', '14F');
```

3 Airport Database DDL

All columns are NOT NULL. Primary keys are defined when the tables are created. The rename, type change, column deletion and foreign keys are added after that.

```
CREATE SCHEMA IF NOT EXISTS lab2_airport;
SET search_path TO lab2_airport;

CREATE TABLE airline_info (
    airline_id INT PRIMARY KEY,
    airline_code VARCHAR(30) NOT NULL,
    airline_name VARCHAR(50) NOT NULL,
    airline_country VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    info VARCHAR(50) NOT NULL
);

CREATE TABLE airport (
    airport_id INT PRIMARY KEY,
    airport_name VARCHAR(50) NOT NULL,
    country VARCHAR(50) NOT NULL,
    state VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

CREATE TABLE baggage_check (
    baggage_check_id INT PRIMARY KEY,
    check_result VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    booking_id INT NOT NULL,
    passenger_id INT NOT NULL
);

CREATE TABLE baggage (
    baggage_id INT PRIMARY KEY,
    weight_in_kg DECIMAL(4,2) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    booking_id INT NOT NULL
);

CREATE TABLE boarding_pass (
    boarding_pass_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    seat VARCHAR(50) NOT NULL,
    boarding_time TIMESTAMP NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

CREATE TABLE booking_flight (
    booking_flight_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    flight_id INT NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

CREATE TABLE booking (
    booking_id INT PRIMARY KEY,
    flight_id INT NOT NULL,
    passenger_id INT NOT NULL,
    booking_platform VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    status VARCHAR(50) NOT NULL,
    price DECIMAL(7,2) NOT NULL
);

CREATE TABLE flights (
    flight_id INT PRIMARY KEY,
```

```

sch_departure_time TIMESTAMP NOT NULL,
sch_arrival_time TIMESTAMP NOT NULL,
departing_airport_id INT NOT NULL,
arriving_airport_id INT NOT NULL,
departing_gate VARCHAR(50) NOT NULL,
arriving_gate VARCHAR(50) NOT NULL,
airline_id INT NOT NULL,
act_departure_time TIMESTAMP NOT NULL,
act_arrival_time TIMESTAMP NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL
);

CREATE TABLE passengers (
    passenger_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    gender VARCHAR(50) NOT NULL,
    country_of_citizenship VARCHAR(50) NOT NULL,
    country_of_residence VARCHAR(50) NOT NULL,
    passport_number VARCHAR(20) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

CREATE TABLE security_check (
    security_check_id INT PRIMARY KEY,
    check_result VARCHAR(20) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    passenger_id INT NOT NULL
);

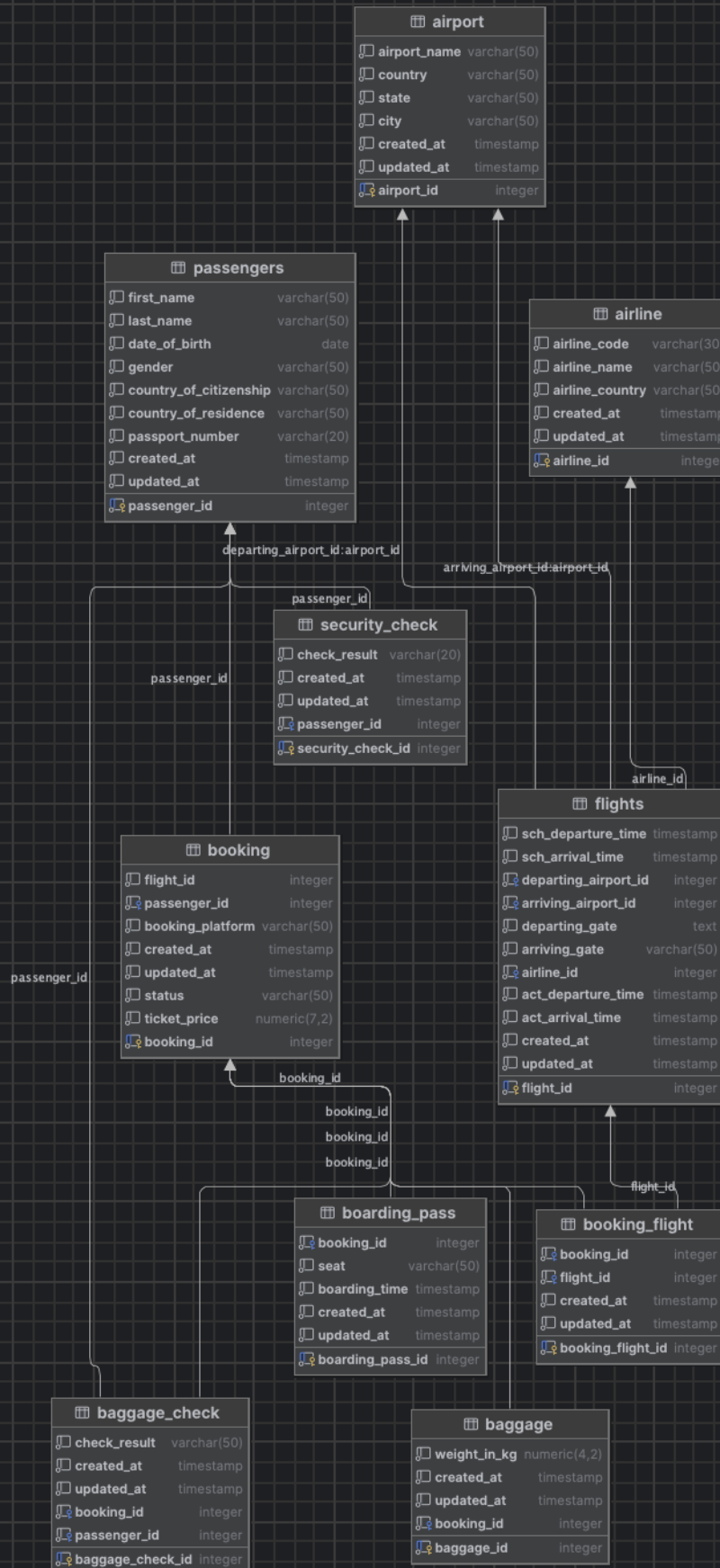
ALTER TABLE airline_info RENAME TO airline;
ALTER TABLE booking RENAME COLUMN price TO ticket_price;
ALTER TABLE flights ALTER COLUMN departing_gate TYPE TEXT;
ALTER TABLE airline DROP COLUMN info;

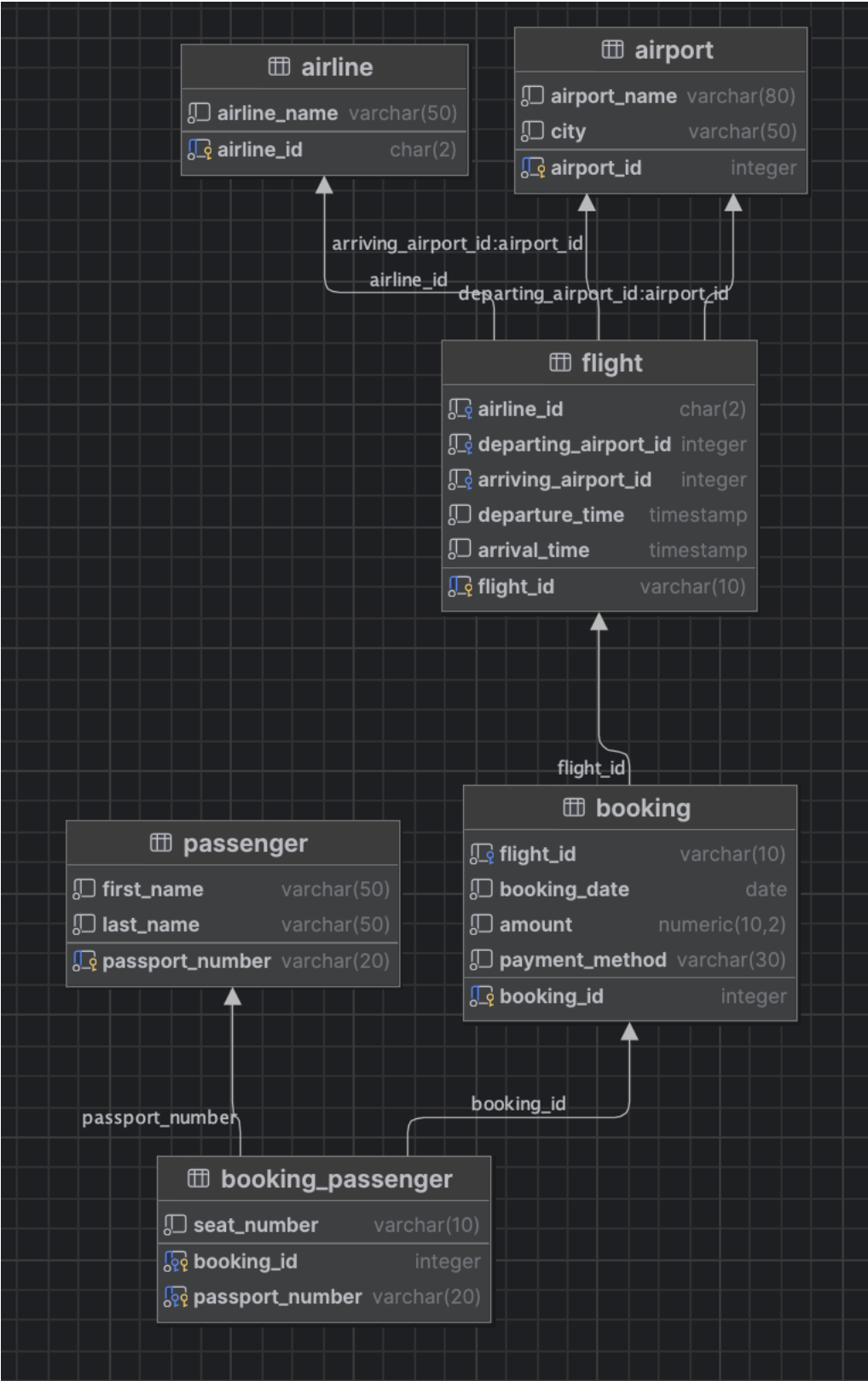
ALTER TABLE security_check
    ADD FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
ALTER TABLE booking
    ADD FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
ALTER TABLE baggage_check
    ADD FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);

ALTER TABLE baggage_check
    ADD FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
ALTER TABLE baggage
    ADD FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
ALTER TABLE boarding_pass
    ADD FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
ALTER TABLE booking_flight
    ADD FOREIGN KEY (booking_id) REFERENCES booking(booking_id);

ALTER TABLE booking_flight
    ADD FOREIGN KEY (flight_id) REFERENCES flights(flight_id);
ALTER TABLE flights
    ADD FOREIGN KEY (departing_airport_id) REFERENCES airport(airport_id);
ALTER TABLE flights
    ADD FOREIGN KEY (arriving_airport_id) REFERENCES airport(airport_id);
ALTER TABLE flights
    ADD FOREIGN KEY (airline_id) REFERENCES airline(airline_id);

```





4 Short Explanation of NOT NULL

NOT NULL means that the column cannot be empty. It is used here because the task requires every attribute to have a value. For example, a passenger must have an ID and a flight must have departure and arrival data.

5 Conclusion

The receipt table was divided into smaller 3NF tables to reduce repeated data. The airport database tables, keys, changes and relationships were also created with PostgreSQL DDL.